Lead 16.1: where kernel saturation pays, the feasibility crossover in m

Round 4 of `RESEARCH_ISOGENY_FIBERS_ECDLP.md` §16.1. Toy Koblitz curves only: `icv1-f2m17-tm101-00378d4e`, `icv1-f2m19-t797-b6cf2467` and `icv1-f2m23-t5197-69e76b73`. No sub-rho claim and no claim about any deployed curve. Full tables, exact commands and limits are in `RESULTS.md` in this directory.

Question and falsifier

The affine law (Prop. 15.2) charges a base closed under the rational 2-torsion kernel $M = \langle T \rangle$ a mitm cost factor $|M|^{m-2}$. The u-frame symmetrised systems have lower degree. Is there a smallest (n, m) where the plain x-frame arm is infeasible under a shared counted cap but the symmetrised arm is not, and does that arm beat the best feasible arm at the same n? **Falsifier:** a crossover arm that beats every feasible arm in total cost.

What ran

- **Panel** (`examples/r4_16_1_m_crossover.rs`): the library oracles with `InheritedF4` (max degree 3) and the same `node_budget = 4096` for every solve. Each (n, m) has one shared $V \ni 1$ of dimension $\ell = \left\lceil \frac{n + \log_2 m!}{m} \right\rceil$, plus invariant-$V$ cells ($\ell = 9$ at $n = 17$, $\ell = 12$ at $n = 23$). There are 24 subgroup targets per curve, and every verdict is gated against exact enumeration (0 gate failures in all 25 cells). The unit is engine-counted word XORs, a lower bound.
- **Anchors** (`ic bench`): end-to-end with a planted logarithm, verified, and 8 matched counted rho runs, on the invariant bases, under the same node budget. The $m = 3$ arms are capped at 300 trials and 10 min of contended wall time.

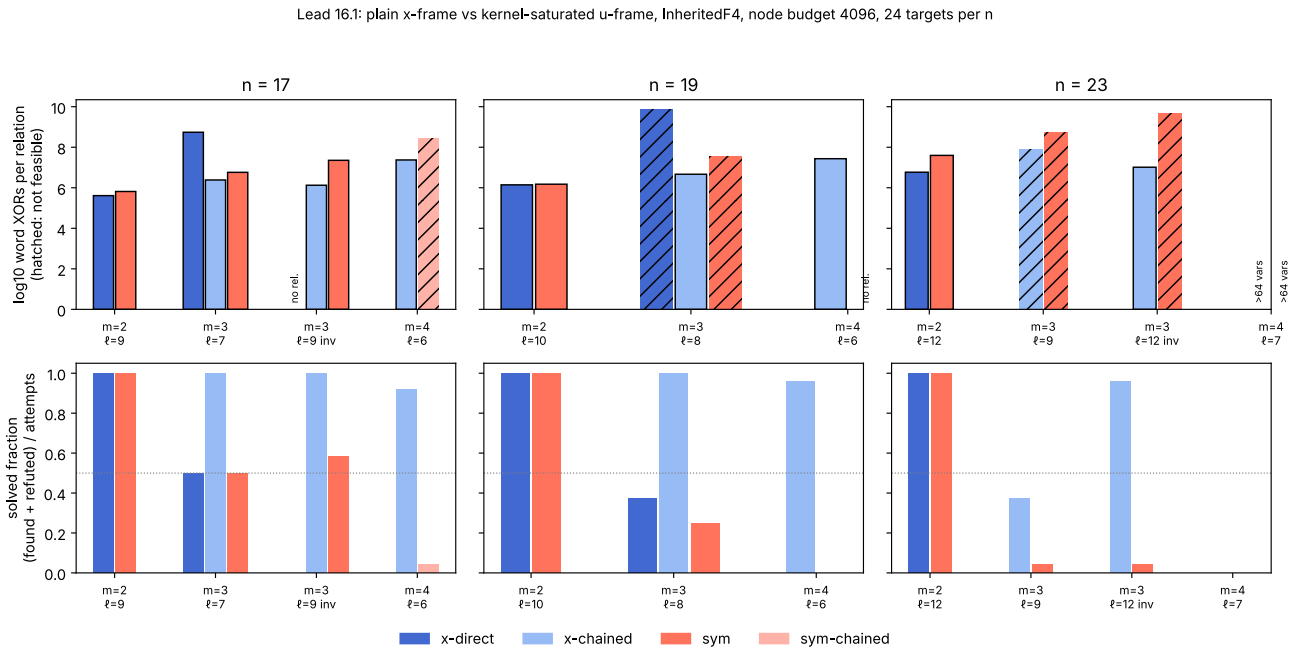


Figure 1: Measured panel, 24 targets per n, node budget 4096. Top: $\log_{10}$ of word XORs per relation found (hatched: not feasible, i.e. fewer than half the attempts solved, or none found). Bottom: solved fraction. The $n = 23$, $m = 3$ `x-direct` cells were not run (see Limits). At $n = 23$, $m = 4$ both chained layouts exceed the engine's 64 unknowns. Data: `crossover_data.csv`, from `results/panel_*.json`.

Key measured numbers

cell	arm	XOR / relation	solved	feasible
$n = 17, m = 3, \ell = 9$ inv	x-direct (S_4, deg 6)	— (1 target, 905 s)	0/1	no
	x-chained	1.33e6	24/24	yes
	sym (S_4, deg 4)	2.26e7	14/24	yes
$n = 19, m = 3, \ell = 8$	x-direct	7.12e9	9/24	no
	x-chained	4.65e6	24/24	yes
	sym	3.45e7	6/24	no
$n = 17, m = 4, \ell = 6$	x-chained	2.36e7	22/24	yes
	sym-chained	2.75e8	1/24	no
$n = 23, m = 3, \ell = 12$ inv	x-chained	1.03e7	23/24	yes
	sym	4.57e9	1/24	no

End-to-end (`ic bench`, $S = \mathrm{GAE}\ /\sqrt{r}$): at $n = 17$, rho 1.76, mitm $m = 2$ 17.1, x $m = 2$ 757, sym $m = 2$ 3 112, **sym $m = 3$ 74 460**, and the direct x $m = 3$ arm gave no output in 10 min. At $n = 23$, rho 1.09, mitm $m = 2$ 143, **sym $m = 3$ 938 966**, and the direct x $m = 3$ arm gave no output in 10 min.

Verdict: confirmed, the falsifier is not met

1. A crossover exists only against the direct S_4 descent. The smallest cell is $(n = 17, m = 3)$ on the invariant $\ell = 9$ base, where the symmetrised arm solves and recovers the logarithm and the direct arm does not.
2. Against the chained x-frame system, which is already in the library, there is no crossover anywhere in $n \in \{17, 19, 23\}$, $m \in \{3, 4\}$. The chain is feasible wherever the symmetrised arm is, and 2.4× to 440× cheaper per relation. At $m = 4$ the kernel-saturated chain is the infeasible one.
3. The crossover arm is 98× (x $m = 2$) and 4 360× (mitm $m = 2$) above the best feasible arms at $n = 17$, and far above rho. Kernel saturation buys feasibility only against one particular plain system, never total cost.

Limits

The x-direct $m = 3$ arm at $n = 23$ was not run in the panel: its first attempt was killed after 30 min inside one solve. Its infeasibility rests on the `ic` anchor (no output in 10 min) and on the 905 s solve at $n = 17$, so it is a wall observation. No direct S_5 exists in either frame, so $m = 4$ compares chains only. With 24 targets per cell, hit rates carry noise of about ±0.1. The panel and `ic` disagree on the sym $m = 3$, $\ell = 12$ hit rate (0.04 against 0.22), which I did not resolve. Word XORs exclude the Macaulay build, and wall times are contended.